

Sachin Nandakumar

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Machine learning engineer and data scientist with 8 years of experience, the last five building production NLP and ML systems at deecoob. I gravitate towards problems that sit at the intersection of language and scale: multilingual text understanding, embedding-based retrieval, and the kind of messy real-world data that makes simple solutions fall apart. Lately that has extended into LLMs, RAG pipelines, and agentic workflows. I am comfortable across the full stack of a modern ML system and have stepped into technical leadership when the team needed it. Open to senior roles in AI Engineering, ML Engineering, Data Science, and MLOps.

EXPERIENCE

Data Scientist · deecoob GmbH

Dresden, Germany · Jun 2021 – present

Promoted from Junior Data Scientist (2022)

Architecture & Infrastructure

- Designed and delivered a scalable platform of 16 Python microservices spanning data ingestion, entity resolution, ML inference, and search. Deployed on GCP with Kubernetes, Helm, and ArgoCD, provisioned via Terraform, processing millions of crawled records daily via Kafka.
- Acted as technical lead through active delivery cycles: set architecture direction, maintained stakeholder commitments, and carried primary codebase ownership through team resizing.
- Maintained and improved CI/CD pipelines (Bitbucket Pipelines) and optimised Docker build configurations, keeping deployments reliable and build times efficient.

Project: Entity Resolution Pipeline

- Owned the core entity resolution pipeline serving 3 production customers across Germany, Poland, and the Netherlands. Handles two problems: deduplication of internally crawled data, and matching customer-data against ours. Precision-recall tuned per client as recall on customer matching is the primary value signal, dependent on clean internal deduplication.
- Built the initial on-prem pipeline using gradient-boosted classification (XGBoost), reaching ~85% deduplication accuracy with precision-recall tuning per client SLA.
- Re-architected for cloud scale as an embedding-based retrieval pipeline: one service generates composite multilingual embeddings via a JinaAI semantic encoder, a second performs fast ANN similarity search against a pgvector index, processing millions of records daily. Delivered measurable improvement in deduplication quality over the on-prem baseline.

Project: Address Enrichment

- Developed an address enrichment pipeline feeding into entity resolution: normalises raw addresses via a self-hosted Nominatim instance on GCP, handling structured, unstructured, and reverse geocoding inputs across country-specific formats.
- Enriched output feeds directly into the entity resolution pipeline, improving match quality across internal and customer data.

Project: Multilingual advertisement classification

- Delivered a multilingual text classification system using fine-tuned transformer models, serving production traffic across English, German, Polish, and Dutch. Covered the full lifecycle: data curation, fine-tuning, serving, and monitoring.
- Integrated into the customer UI as a live feature, creating a continuous feedback loop where user corrections feed back into retraining cycles to improve model quality over time.

Internship, Master Thesis · 100 Worte Sprachanalyse GmbH

Heilbronn, Germany · Oct 2020 – Jul 2021

- Thesis: Exploring Authenticity Modulation in Texts using Seq2Seq Transformer Models (graded 1.1). Fine-tuned BART and T5 with control codes for paraphrasing and bidirectional psychological style transfer.
- Engineered a 7-million-entry parallel paraphrase corpus combining web-scraped corporate content and public datasets, using back-translation via Chinese as pivot language; designed and ran a human annotation pipeline for evaluation.

Test Engineer, R&D IVS · Infosys Ltd.

Bengaluru, India · Jun 2016 – Aug 2018

- Developed ML models for autonomous e-commerce testing, including a Q-learning (reinforcement learning) test-case prioritisation system predicting execution relevance.
- Won the InStep Mentor Award for exceptional mentorship during the Infosys Global Internship Program; delivered a tech talk on reinforcement learning in autonomous testing.

PERSONAL PROJECTS

linguaalayam: Malayalam Dictionary RAG Pipeline

- RAG pipeline over the Olam EN→ML corpus (~58k entries) with hybrid retrieval: exact headword match, trigram fuzzy search (pg_trgm), and HNSW cosine semantic search via pgvector, orchestrated through a LangGraph pipeline with query understanding, cross-encoder reranking, and LLM answer synthesis via LiteLLM.
- Exposed as an MCP server, making the dictionary directly pluggable as a tool into any MCP-compatible LLM client like Claude desktop.
- Offline evaluation harness measuring Hit@k and MRR across a labelled query set; CI/CD via GitHub Actions, pre-commit hooks, Alembic migrations, Poetry, and semantic versioning.

readme-drift: README Drift Detection for Pre-commit and CI

- Python CLI tool and pre-commit hook that detects stale README references after code changes using AST-based diff analysis, catching renamed symbols, removed functions, and shifted file paths before they silently break documentation.
- Published to PyPI; configurable via pyproject.toml, integrates into GitHub Actions CI and pre-commit pipelines, with semantic versioning via Commitizen and Poetry.

job-ledger: End-to-end job intelligence pipeline

- Apify scraping across multiple job boards, Claude-powered CV-to-JD match scoring, Streamlit UI with live background runs, SQLAlchemy/SQLite persistence. Processes 200+ listings per run with full deduplication, caching, and incremental DB writes. Developed using Claude Code.

EDUCATION

M.Sc. Data and Knowledge Engineering

Otto-von-Guericke-Universität Magdeburg

· GPA 1.7

· 2018 – 2021

B.Tech. Information Technology

Viswajyothi College of Engineering & Technology, Kerala

· GPA 2.1

· 2012 – 2016

SKILLS & EXPERTISE

Large Language Model & Agentic AI: RAG pipelines, LangGraph, FastMCP, LiteLLM, LLM integration, Anthropic SDK, Apify, agentic workflows, prompt engineering

NLP & ML: HuggingFace Transformers, SentenceTransformers, TensorFlow, PyTorch, Scikit-learn, XGBoost, SpaCy, Optuna, TFDV, fine-tuning, model evaluation, hybrid retrieval, vector search, cross-encoder reranking

MLOps: Docker, Kubernetes, Helm, ArgoCD, Terraform, GitHub Actions, Bitbucket Pipelines, Data Versioning Control (DVC), CI/CD

Data & Search: PostgreSQL, SQL, pgvector, Alembic, Supabase, Elasticsearch, Apache Kafka

Cloud: Google Cloud Platform (GCP)

Software Development: Python, FastAPI, Streamlit, REST APIs, gRPC, Protocol Buffers, SQLAlchemy, Hydra, Poetry, Pytest, pre-commit, semantic versioning, clean code & architecture

Analytics: Matplotlib, Plotly, Kibana

Languages: English (fluent), German (basic)